

Assignment 1

Objective

The objective of this assignment is to apply text preprocessing techniques and build a machine learning model to classify news articles as **Fake** or **Real**.

Dataset

Use the following dataset [Fake News Dataset](#)

Tasks

Data Exploration Perform exploratory data analysis by examining the dataset structure, checking for missing values, identifying any inconsistencies or anomalies in the data, and Show the distribution of Fake vs Real news.

Text Preprocessing Apply appropriate text preprocessing steps (such as cleaning, tokenization, stopword removal, and stemming or lemmatization, etc.) to prepare the news text for machine learning.

Feature Extraction Apply feature extraction techniques such as Bag of Words and TF-IDF to transform the text data into numerical feature representations suitable for machine learning models.

Model Building Train at least two classification models on the dataset.

Model Evaluation Evaluate the trained models using performance metrics such as Accuracy and F1-score. Compare the performance of the models and select the best-performing model

Teams& Registration

Students must work in teams, and each team should consist of 5 to 6 members.

Team registration must be completed before the deadline. The registration form will close on 31 March. [Registration Form](#)

Deadline& Delivery:

The task submission and discussion will be conducted during lab sessions starting from **April 11**.

Each team should send **one or two representatives** to present and discuss the work with the assigned Teaching Assistant.